

Malpighi's Forgotten Drawings of Renal Glomeruli

Roberto F. Nicosia and Paolo Cassoli



Marcello Malpighi (1628-1694) is universally recognized for his pioneering work in microscopic anatomy. His groundbreaking studies of human, animal, and plant anatomy laid the foundation for modern histology, embryology, comparative anatomy, entomology, botany, and biology. Among his many contributions is the discovery of renal glomeruli, which he described as *minimae glandulae* (very small glands), attributing to them the production of urine. His treatise *De renibus* contained a detailed description of glomeruli but no images. A drawing of glomeruli by Malpighi appeared many years later in an illustrated manuscript sheet containing a plate with different types of glands, including the kidney's *minimae glandulae*. While examining Malpighi's handwritten scientific diary, kept at the Biblioteca Universitaria di Bologna, we found unpublished drawings of his kidney observations, including one that may represent his earliest extant depiction of glomeruli. We review Malpighi's groundbreaking discovery of glomeruli, focusing on his kidney drawings and their historical significance.

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Marcello Malpighi (1628-1694), physician and natural philosopher, was among a small group of 17th-century scientists who first used the microscope to explore the anatomical fabric of life.¹ His groundbreaking work, *De pulmonibus*, published in 1661, revealed for the first time the alveolar architecture of the lungs and the existence of a network of small vessels (capillaries) coating the alveoli.² This landmark study upended the ancient belief that the lungs were made of coagulated blood poured between vessels and fibers and completed William Harvey's discovery of the circulation of blood by revealing the microscopic vasculature that connected arteries to veins. At the suggestion of his mentor Alfonso Borelli (1608-1679), Malpighi illustrated his groundbreaking discoveries with images of pulmonary alveoli and capillaries, which represent the first published drawings of an animal internal organ examined with a microscope.³

After his pioneering work on the lungs, Malpighi conducted numerous studies of organ anatomy in mammals, birds, amphibians, fish, insects, and plants, laying the foundation for modern histology, embryology, comparative anatomy, entomology, botany, and biology. In many of his works, Malpighi documented key anatomical findings with illustrations, which he meticulously drew himself.¹⁻³

Among Malpighi's most important contributions is the discovery of glomeruli. This fundamental observation was published in 1666 in a work titled *De viscerum structura*. This work included 3 treatises on the liver (*De hepate*), cerebral cortex (*De cerebri cortice*), and kidneys (*De renibus*). The definitive version of *De viscerum structura*, which included 2 additional treatises on the spleen (*De liene*) and heart polyp (*De polypo cordis*), appeared in 1668.^{4,5} Notably, *De viscerum structura* was published in haste by Malpighi and surprisingly with no illustrations, likely to ensure he was the first to report these observations.

Malpighi regularly recorded his observations in a scientific diary, which is currently housed at the Biblioteca Universitaria di Bologna.⁶ While examining his diary, we found notes containing drawings of kidneys, including one with a detailed depiction of glomeruli, sketched by the

great anatomist around the time *De viscerum structura* was published. We present these previously unpublished drawings, review the accompanying text, and relate these observations to Malpighi's original description of his landmark discovery in *De renibus*.

The Discovery of Glomeruli

In the introduction to *De renibus*, Malpighi wrote, "I did not find the structure of kidneys with the help of books but with the patient, continuous, and varied use of the microscope" ("me hanc renum structuram librorum ministerio nequaquam assecutum, sed patienti, longo, & vario usu microscopii").⁴ From Malpighi's writings, we know that he used 2 types of instruments: the "flea glass" (*lens pulicaris*), a single-lens instrument with relatively low magnification, and the compound microscope, a 2-lens instrument (*microscopium duarum lentium*) with much higher magnification and a smaller view field.² At least some of his instruments were manufactured by Eustachio Divini (1610-1685), a renowned Italian lens maker whose compound microscopes could magnify up to 143 times.⁷

When Malpighi examined the kidneys of quadrupeds, turtles, and humans under a microscope, he recognized numerous small channels, which he described as tubules, canaliculi, or excretory vessels, confirming the observation made a few years earlier by his younger colleague Lorenzo Bellini (1643-1704)⁸ and anticipated in the 16th century by Bartolomeo Eustachio (circa 1500/1510-1574).⁹ Malpighi found that tubules could be visualized even better by spreading ink with a finger over the external surface of the cortex or its cut surface. The ink bound to the interstitium, enhancing the visibility of the tubules which remained unstained.^{4,10}

The ink-staining method also revealed, in the external part of the kidney (cortex) and among the tubules, the glomeruli, which appeared to Malpighi as "innumerable small globules" resembling "fish eggs." He described them as "very small glands" (*minimae glandulae*) and attributed to them the "secretion" of urine. He then found that these

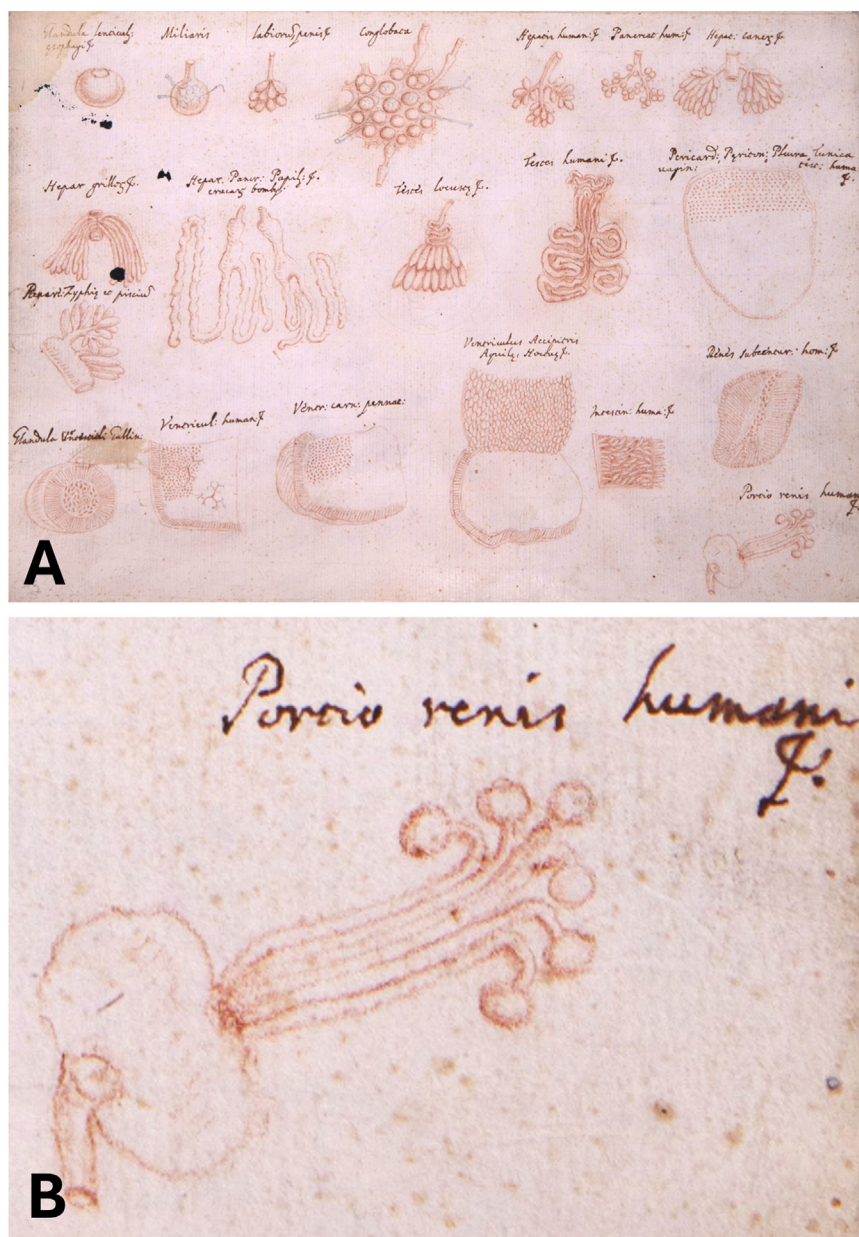


Figure 1. Manuscript sheet with Malpighi's drawings of glands (ms 936, II, A, c. 21, Biblioteca Universitaria di Bologna). (A) Complete view of the drawings of 20 glands, including the kidney (bottom right). Glands depicted on this sheet were described by Malpighi in *De structura glandularum conglobatarum consimiliumque partium*, published in 1689. (B) Close-up view of the kidney drawing, titled *Portio renis humani* (part of the human kidney), showing 6 glomeruli and attached tubules arising from a kidney. Reproduced with permission of the copyright holder (Alma Mater Studiorum Università di Bologna, Biblioteca Universitaria di Bologna).

“glands” could be further highlighted by injecting the renal artery with a black liquid mixed with distilled spirit (*spiritus vini*). The black fluid stained the arterial vascular tree and its finest cortical branches as well as the *minimae glandulae*, demonstrating a connection between the arterial circulation and the glomeruli. After stripping the fibrous capsule, Malpighi could see the blackened glomeruli attached to the fine ramifications of the arterial tree, like apples to tree branches.^{4,10}

Malpighi observed that only part of the glomerulus blackened with ink while the rest maintained its own color.

He reasoned that if it were true that urine originated from arterial blood (“*si verum ... urinae materiam ab arteriis derivari*”) and arteries ended in the glands, with urine being expelled into the pelvis through excretory vessels (papillary ducts), there must be a connection between the glands and tubules. Convinced that the renal glands were directly involved in the secretion of urine into tubules, Malpighi tried repeatedly to demonstrate a connection between the two. The black fluid injected through the arterial circulation swelled the glomeruli but never penetrated into the tubules. He obtained the same negative result by injecting the ureters or

the renal vein.* He even performed vivisection experiments, ligating the ureter and vein of dog kidneys, but never succeeding in demonstrating the connection he was seeking.^{4,10} Malpighi's hypothesis would be fortuitously proven true 176 years later when William Bowman demonstrated that glomeruli and tubules were indeed connected by injecting into the arterial circulation a potassium dichromate/lead acetate solution that penetrated into the tubules after disrupting the integrity of the glomerular capillary walls.¹¹

Malpighi's Previously Published Drawing of Glomeruli

Although Malpighi described his anatomic observations in great detail in *De renibus*, he did not include drawings as he had previously done in *De pulmonibus* for the lung alveoli and blood capillaries. He revisited his observations on the structure of the kidney in a later work entitled *De structura glandularum conglobatarum consimiliumque partium*, which provided a comprehensive study of glands, from the simplest to the most complex examples, including the kidneys.¹² However, even in this later work Malpighi did not include drawings.

The only drawing of glomeruli by Malpighi that, to our knowledge, is cited in the modern literature appears in an illustrated manuscript sheet crafted in preparation for *De structura glandularum*, published in 1689.^{13,14} This manuscript is kept, along with other manuscripts autographed by Malpighi, at the Biblioteca Universitaria di Bologna. The drawing, entitled *Portio renis humani*, shows the outline of a kidney from which arise 6 magnified nephrons, each composed of a glomerulus and tubule. Besides the kidney and glomeruli, the manuscript sheet contains drawings of 19 additional "glands," including simple and complex (conglobate) glands such as the liver, pancreas, and adrenal gland (Fig 1), as well as text with a classification of glands into 4 separate species.

The Unpublished Drawings of Glomeruli in Malpighi's Diary

From 1660 until his death in 1694, Malpighi kept a scientific diary handwritten in Latin, where each new finding was marked by "Observavi" (I observed).⁶ His notes were often accompanied by drawings, remarkable for their accuracy and clarity. Malpighi's diary was discovered in 1830 by his biographer Gaetano Atti in the house of a physician in Crevalcore, Malpighi's birthplace near Bologna, and it was donated to the Biblioteca Universitaria di Bologna where it is currently housed.^{6,15} The diary, titled *Marcelli Malpighi observationes anatomicae autographae in plantis et animalibus*, contains notes from Malpighi's plant and animal studies, including many

dissections performed on various animal species.⁶ Much of this material has not been translated or published.

We recently examined *Observationes anatomicae* and came across 3 notes on mammalian kidneys by Malpighi accompanied by his drawings. The first observation, of a horse kidney, is from June 4, 1667, the year after the publication of *De renibus*. The second, also on a horse kidney, is not dated, but it was likely made in the summer of 1668 because the other notes written by Malpighi before and after this one in this section of the diary were dated June to September 1668. The third, on a dormouse kidney, was written on September 30, 1688. We provide a description of Malpighi's unpublished drawings and the accompanying notes, followed by our interpretation of their significance.

The Horse Kidney

The first observation from 1667 made by Malpighi, with his friends and collaborators Carlo Fracassati (circa 1630-1672) and Silvestro Bonfigliuoli (1637-1696), contains a small drawing depicting the bean-shaped outline of a horse kidney (not shown) and descriptions of the renal arteries, veins, nerves, papillae, tubules, and glomeruli. Although this note lacks drawings of glomeruli, it is particularly significant because it contains Malpighi's description of the injection method he used to identify them, as he had previously reported in *De renibus*.†

In the second note on the horse kidney, Malpighi focused on the renal medulla and excretory apparatus and included a detailed drawing of glomeruli.‡ He wrote that

†"Die 4 iunii 1667 Bononiae, cum domino Fracassato et Bonfiolo in equo haec observavi: Primo renes sunt huius figurae [small drawing of kidney]; ingentes habent nervos insignes subintrantes cum emulgentibus, interius habent papillas sicuti in sue etc. observavi, vasa urinae manifestissima ap[er]uere in fasciculos collecta stipata vasis sanguineis et in parte exteriori reflectebantur intro evidenter et applicabantur glandulis quae rubrae sunt et circumductis vasis sanguineis et injecto atramento per arterias denigrabantur in exteriori superficie vasa sanguinea emergentia hinc inde ramificantur et glandulis continuantur quae copiosissimae sunt et abunde patent." (Biblioteca Universitaria di Bologna, ms. 2085, II, c. 16 v.) Translation: On June 4, 1667 Bologna, with Lord Fracassati and Bonfigliuoli, I observed the following in the horse: First, the kidneys are of this shape [small drawing of kidney]; they have prominent, notable nerves entering [the kidney] with the veins, and internally they have the same papillae as pigs, etc. I observed the urinary vessels very clearly defined, collected into bundles, packed with blood vessels, and on the outer part [the cortex], they were evidently reflected inward and attached to the glands, which are red and surrounded by blood vessels. When ink was injected through the arteries, the blood vessels on the outer surface were darkened. The blood vessels emerging here and there branched out and continued into the glands, which are very numerous and abundantly evident.

‡"Observavi renes equinos qui habent loco papillarum zonam semilunarem a cuius medio effluit urina. Observavi quod intra incisus stilus hinc inde penetrat per ductum qui centrum huius zonae percurrit et in hoc sunt irregularia ora per quae elabitur urina. A ductus in quem fibrae plures desinunt et hic hiat in B. Observavi iterum glandulas renum." (Text of Figure 2: Biblioteca Universitaria di Bologna, ms. 2085, II, c. 30 r.) Translation: I observed the horse kidneys which have, instead of papillae, a semilunar zone [medullary crest] from the middle of which urine flows. I observed that a stylus [inserted] inside a cut penetrates from one side to the other through a duct that runs through the center of this zone, and in this there are irregular openings through which the urine escapes. A: duct into which many fibers [papillary ducts] end, and this opens in B [pelvis]. I observed the kidney glands again.

*"Diu laboravi, ut oculis evidentem hanc continuationem subijcerem, quam ratio sufficienter attingebat, nam infusi liquores per arterias, etsi glandulas repleant, non tamen urinae vasa penetrantes unquam observare potui, quod etiam de venis mihi accidit." (Translation: I labored long to present to the eyes this evident connection [between glomeruli and tubules], sufficiently grasped by reason, but having injected liquids through the arteries, although they swelled the glands, I could never observe them penetrating into the urinary vessels, which also occurred to me with the veins.)

in the horse kidney the urine flows from a crescent-shaped area (*zona semilunarem*), in place of the papillae (*loco papillarum*), into the pelvis through irregular openings within a *ductus*, a tubular structure that he could probe with a pointed instrument (*stilus*). Next to the text, he drew the horse kidney, appropriately labeled to identify the anatomic structures described in his writing. Separately, he wrote, “I have observed the kidney glands again” (“*Observavi iterum glandulas renum*”), likely referring to his earlier observation from the previous year, and he drew 2 sets of magnified glomeruli and tubules (Fig 2).

To fully understand this note, a brief overview of the horse kidney anatomy is needed. Unlike the human kidney, which is multilobar and has multiple medullary pyramids and papillae, the horse kidney is unilobar with a single broad medullary region and a continuous ridge, known as the medullary crest, formed by the fusion of multiple renal papillae. The horse kidney’s excretory apparatus is also significantly distinct. Whereas the human kidney has a complex system of minor and major calyces that open into the pelvis, the horse kidney’s excretory apparatus consists of a central cavity connected to 2 tubelike recesses that drain the urine from the poles of the kidneys. This structural difference accommodates the larger volumes of urine produced by the horse.¹⁶ Thus, the crescent-shaped area observed by Malpighi corresponded to the medullary crest of the horse kidney, and what he referred to as a “*ductus*” (labeled A in his drawing) was the tubular recesses of the pelvis.

The attention Malpighi paid to the anatomy of the horse kidney in his diary is not surprising, as he relied heavily on comparative anatomy to understand organ structure and function. In the first chapter of *De renibus* he described the morphology of kidneys from various species but made no mention of the horse.^{4,10} Thus, he must have dissected the horse kidney after publishing *De renibus* and decided to annotate the new findings in his diary.

Of special interest in this note are the 2 drawings of glomeruli with attached tubules. Because no illustrations were included in *De renibus*, this drawing, sketched by Malpighi 2 years after the treatise was first printed, may represent the first extant depiction of glomeruli by the great anatomist that is available to us. Intriguingly, the glomeruli are shown in 2 separate bundles, with tubules converging medially, particularly in one of the drawings. Why did Malpighi group the glomeruli into a bundle? The answer lies in Malpighi’s interpretation of the anatomical structure of kidneys.

When describing the gross anatomy of kidneys in *De renibus*, Malpighi explained that each kidney is subdivided into smaller parts, which in some species form separate lobes. He wrote that subdivisions are also evident in the human kidney, and, though the lobar morphology tends to disappear in adults, they persist internally as demonstrated by the multiple medullary

pyramids. He applied the concept of subdivisions also to the glomeruli, extending it to the microscopic level. Glomeruli, he wrote, formed groups of more than 40 in association with bundles of tubules from which emerge the smallest subdivisions of the kidney.^{4,10} Here he was likely referring to the glomeruli located between medullary rays (the straight portions of the proximal and distal tubules and the collecting ducts), which extend from the medulla into the cortex. Notably, Malpighi sketched the medullary rays with straight lines in the horse kidney drawing without describing them (Fig 2).

Although Malpighi had been unable to prove the existence of a connection between glomeruli, tubules, collecting ducts, and papillary ducts, he was correctly envisioning the microscopic organization of the nephron and its draining system. By depicting many “renal glands” together, with their respective tubules forming a common bundle, he was illustrating the smallest subdivisions and structural units of the kidney responsible for urine production.

The Dormouse Kidney

In 1688, the year before *De structura glandularum* was published, Malpighi dissected a dormouse, a small rodent of the family *Gliridae*, and recorded in his diary his observations on the animal’s urogenital system, stomach, and liver (Fig 3).⁸ After describing the dormouse’s bicornuate uterus, ovaries, vagina, external genitalia, and bladder, Malpighi turned his attention to the kidneys, which he portrayed as *elegantes*, likely referring to what he felt was their refined anatomy. He wrote that the kidneys have numerous red glands (*glandulas copiosas rubras*), located at their periphery, as well as whitish tubules that ended at the papilla in the middle of the kidney.

Sketched next to Malpighi’s writing is the drawing of a small kidney (Fig 3). The drawing highlights the anatomy of the dormouse kidney, focusing on the

§“30 eiusdem (septembris 1688) Observavi in glire foemina uterum qui oblongis tubis constat quarum extremitatibus globuli ovals locabuntur coagmentati vesiculis seu ovis minimis. Tubae desinebant in corpus uteri quod elongabatur in vaginam, in quam vesica urinaria suo ductu desinebat, et prope pudendum foemineum intus subalbum locabatur corpus quasi glandulosum. Observavi pariter renes, qui elegantes sunt; habent enim glandulas copiosas rubras in periferia vasa quoque excretoria subalba quae ad papillam desinunt in medio renis in pelvi. In tunica interiori ventriculi observavi fistulas. In iecore lobuli integrantur minimis utriculis.” (Text of Figure 3: Biblioteca Universitaria di Bologna, ms. 2085, II, c. 118 r.) Translation: On September 30, 1688, I observed in a female dormouse a uterus consisting of elongated tubes, at the ends of which oval globules [ovaries] were located, containing small vesicles or eggs. The tubes ended in the body of the uterus, which extended into the vagina, into which the duct of the urinary bladder terminated, and near the female genitalia, there was a whitish and almost glandular body located internally. I also observed the kidneys, which are elegant; for they have numerous red glands on the periphery and also whitish excretory vessels that end at the papilla in the middle of the kidney in the pelvis. In the inner lining of the stomach, I observed fistulas [small ducts?]. In the liver, the lobules are integrated with very small vesicles [bile ducts?].

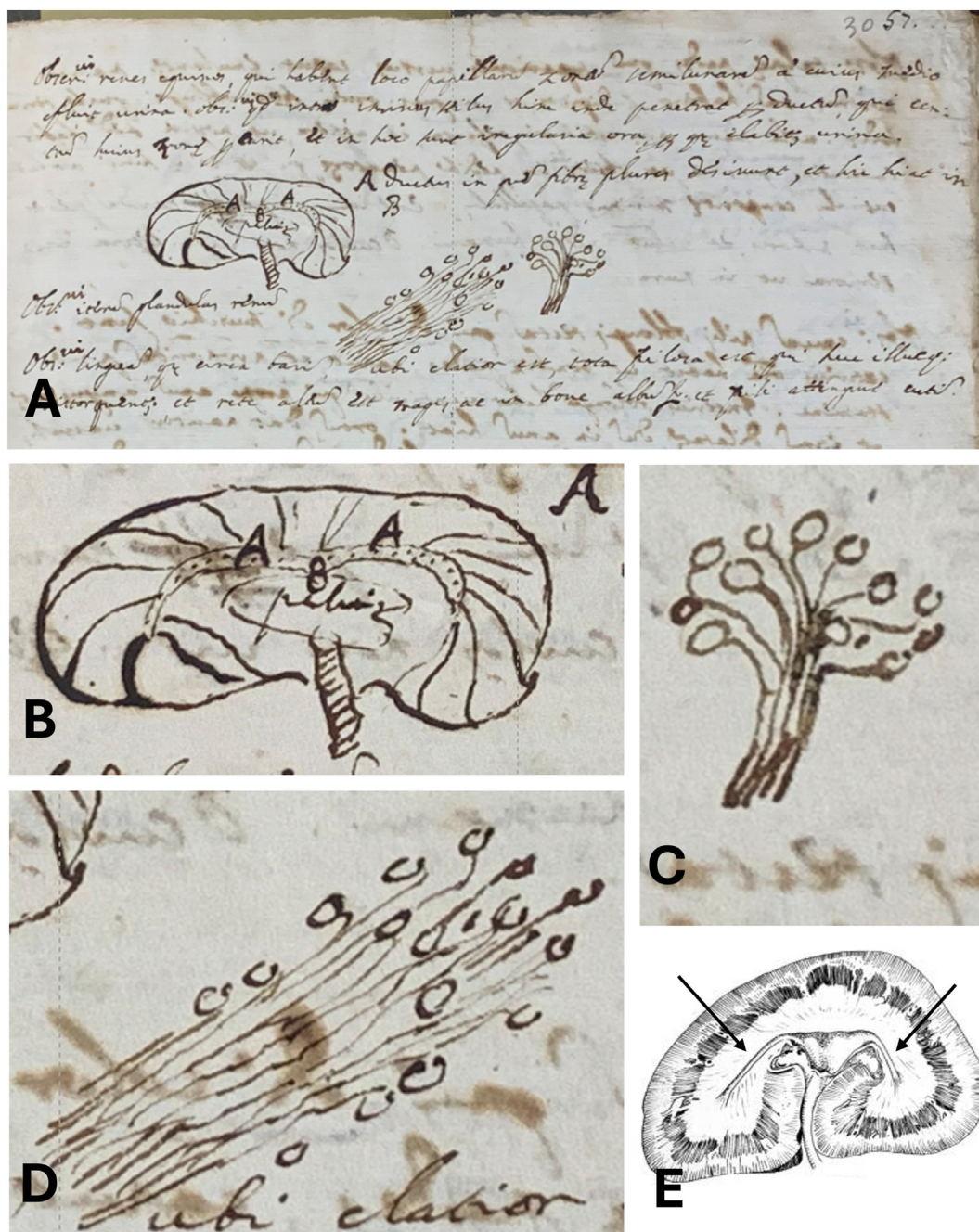


Figure 2. Note on the horse kidney from Mapighi's scientific diary (*Observationes anatomicae autographae in plantis et animalibus*, ms. 2085, II, c. 30 r., *Biblioteca Universitaria di Bologna*). (A) Malpighi's description and drawing of the horse kidney, with separate drawings of glomeruli and tubules. (B) Drawing of the horse kidney gross anatomy: Malpighi labeled "A" the "*ductus*," into which urine flows and "B" the region where it opens into the "*pelvis*." (C) Drawing of glomeruli and tubules. (D) Additional drawing of glomeruli and tubules. (E) Schematic of horse kidney anatomy in which the tubular recesses of the excretory apparatus, marked by arrows, correspond with the "*ductus*" labeled "A" in Malpighi's drawing. A-D: Reproduced with permission of the copyright holder (Alma Mater Studiorum Università di Bologna, Biblioteca Universitaria di Bologna). E: Reproduced with permission of the copyright holder (Charles Allen Cary Veterinary Medical Library at Auburn University Libraries, <https://content.lib.auburn.edu/digital/collection/vetmed1/id/255/>).

location of glomeruli and tubules. The glomeruli are drawn as dots in the cortical region. Although the glomeruli are not sketched separately and in detail, this drawing is remarkable for the attention Malpighi paid to

the unipapillary anatomy of the dormouse kidney, which resembles the anatomy of the mouse kidney.¹⁷ Malpighi depicted the tubules as straight or slightly wavy lines, likely emphasizing the medullary rays as he



Figure 3. Note on the dormouse kidney from Malpighi's scientific diary (*Observationes anatomicae autographae in plantis et animalibus*, ms. 2085, II, c.118 r.). (A) Malpighi's description of the dormouse urogenital system, stomach and liver. The fifth and sixth lines describe the anatomy of the kidney and are accompanied by a drawing. (B) Malpighi's depiction of the dormouse kidney includes glomeruli, drawn as dots in the renal cortex. (C) Schematic of unilobar/unipapillary kidney anatomy. A-B: Reproduced with permission of the copyright holder (Alma Mater Studiorum Università di Bologna, Biblioteca Universitaria di Bologna). C: Reproduced with permission of the copyright holder (Clinical Tree, <https://veteriankey.com/urinary-system-2/>).

had done many years earlier for the horse kidney drawing.

Conclusion

We have presented here 2 previously unpublished drawings of kidneys sketched by Marcello Malpighi in his scientific diary, where he regularly noted his anatomical observations. These drawings carry historical significance because they are rare examples of Malpighi's depiction of kidney anatomy. Particularly noteworthy are those of the horse glomeruli, which were likely crafted around the time *De renibus*—Malpighi's seminal work on the kidneys—was published. Malpighi's artwork not only highlights his pioneering contribution to microscopic anatomy but is also a testament to his passion for drawing, which drove him to reproduce on paper many of his pioneering observations and discoveries.

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